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MAHATMAGANDHIUNIVERSITY, KOTTAYAM
MGU-UGP(HONOURS)REGULAREXAMINATIONMarch2026

FourthSEMESTER

SkillEnhancementCourse(SEC)-MG4SECFTQ200-BAKING TECHNOLOGY

(2024ADMISSIONONWARDS)

Duration:1Hour30Minutes **MaximumMarks:50**

PartA

ShortAnswer type
*Answer any **5** questions*
*Each question carries **2** marks*

1. Name two cleaning agents used in bakeries. (CO1)

Detergents and sanitizers (such as chlorine-based sanitizers or quaternary ammonium compounds).

2. What is compressed yeast? (CO2)

Compressed yeast is a moist, solid form of baker's yeast containing about 70% moisture, used as a biological leavening agent in bread fermentation.

3. Define bread staling. (CO6)

Bread staling is the process in which bread loses its freshness and softness over time due to starch retrogradation and moisture redistribution.

4. Define rope disease in bread. (CO3)

Rope disease is a bacterial spoilage of bread caused mainly by Bacillus species, resulting in sticky, stringy crumb and a fruity odor.

5. What is over-mixing of dough? (CO2)

Over-mixing of dough occurs when mixing continues beyond optimal gluten development, causing gluten breakdown and poor dough structure.

6. What is creaming method? (CO6)

The creaming method is a cake-mixing technique in which fat and sugar are beaten together to incorporate air before adding eggs and flour.

7. Outline the steps in cracker production. (CO4)

Mixing of ingredients → dough resting/fermentation → sheeting → cutting → baking → cooling → packaging.

Part B

Short Essay Questions

Answer any 5 questions

Each question carries 4 marks

8. Explain the manufacturing method of sponge cake using a flow diagram. (CO6)

Egg beating → Addition of sugar → Folding of flour → Batter preparation → Pouring into moulds → Baking → Cooling → Packaging.

9. How would enzyme activity during fermentation influence bread quality, volume and texture? (CO2)

During fermentation, enzymes such as amylases convert starch into fermentable sugars. Yeast uses these sugars to produce carbon dioxide and alcohol. The CO_2 gas expands the dough, increasing bread volume. Enzyme activity also improves crumb texture, softness, flavor and crust color.

10. Explain the manufacturing method of cookies using flow diagram. (CO6)

Weighing ingredients → Creaming fat and sugar → Addition of eggs and flavor → Mixing with flour → Dough preparation → Shaping/moulding → Baking → Cooling → Packaging.

11. Write a detailed note on non-fermented baked foods. (CO1)

Non-fermented baked foods are products that do not undergo yeast fermentation. Instead, they may use chemical leavening agents or no leavening agents at all. Examples include biscuits, cookies, crackers and certain cakes. These products rely on mixing, shaping and

baking processes to obtain their texture and structure.

12. Describe the quality characteristics of good biscuits and crackers and how they are evaluated. (CO4)

Good biscuits and crackers should have uniform shape, attractive color, pleasant aroma, crisp texture and proper moisture content. Evaluation is done using sensory evaluation (taste, aroma, appearance, texture), physical tests (thickness, spread ratio), and chemical tests (moisture and fat content).

13. Explain the preparation of angel food cake. (CO5)

Angel food cake is prepared mainly using egg whites. Egg whites are whipped to form foam, sugar is gradually added, and flour is gently folded into the mixture. The batter is poured into moulds and baked. The cake is cooled upside down to maintain its light and airy structure.

14. Explain pH value of flour. (CO2)

The pH of flour generally ranges between 6.0 and 6.8, which is slightly acidic. This pH influences enzyme activity, yeast fermentation and dough properties. Proper pH helps maintain gluten strength and bread quality.

PartC

Long Essay Questions

*Answer any **2** questions*

*Each question carries **10** marks*

15. Explain time and temperature requirements for common baked products. (CO1)

Different baked products require specific baking temperatures and times. Bread is typically baked at 200–230°C for 25–35 minutes. Cakes are baked at 160–180°C, while biscuits and cookies are usually baked at 180–220°C for shorter durations. Proper control of time and temperature ensures good volume, crust color, texture and product quality.

16. Define leavening agents and explain their classification. (CO2)

Leavening agents are substances that produce gas in dough or batter, causing it to expand and become light and porous. They are classified into:

- Biological leavening agents: Yeast
- Chemical leavening agents: Baking powder, baking soda
- Physical leavening agents: Air and steam.

17. Define fermentation in bread making and factors affecting it. (CO3)

Fermentation in bread making is the process by which yeast converts sugars into carbon dioxide and alcohol, causing the dough to rise. Fermentation occurs during dough resting and proofing stages. Factors affecting fermentation include temperature, yeast quantity, sugar concentration, pH, dough consistency and fermentation time.

18. Describe different types of biscuits and compare their manufacturing methods. (CO6)

Biscuits are commonly classified into soft dough biscuits, hard dough biscuits and fermented biscuits.Soft dough biscuits contain higher fat and sugar and require less gluten development. Hard dough biscuits have lower fat and sugar and involve stronger gluten formation. Fermented biscuits undergo dough fermentation before baking. Their manufacturing methods differ mainly in dough preparation, fermentation and sheeting processes.